

Chapter 33

Random Number Generator (RNG)

33.1 Introduction

The ESP32-P4 contains a true random number generator, which generates 32-bit random numbers that can be used for cryptographical operations, among other things.

33.2 Feature List

The random number generator in ESP32-P4 generates true random numbers, which means random numbers generated from a physical process, rather than by means of an algorithm. No number generated within the specified range is more or less likely to appear than any other number.

33.3 Functional Description

Every 32-bit value that the system reads from the [LPSYSREG_RNG_DATA_REG](#) register of the random number generator is a true random number. These true random numbers are generated based on the system's **thermal noise**, the **asynchronous clock mismatch** and the **BUF_CHAIN**.

- **Thermal noise** comes from the SAR ADC. Whenever the SAR ADC is enabled, bit streams will be generated and fed into the random number generator through an XOR logic gate as random seeds.
- RC_FAST_CLK is an **asynchronous clock** source and it increases the RNG entropy by introducing circuit metastability. Please refer to the [Note](#) about the RC_FAST_CLK.
- The count value generated by the **BUF_CHAIN**, implemented using BUF_CELL, is used as a random seed for the random number generator.

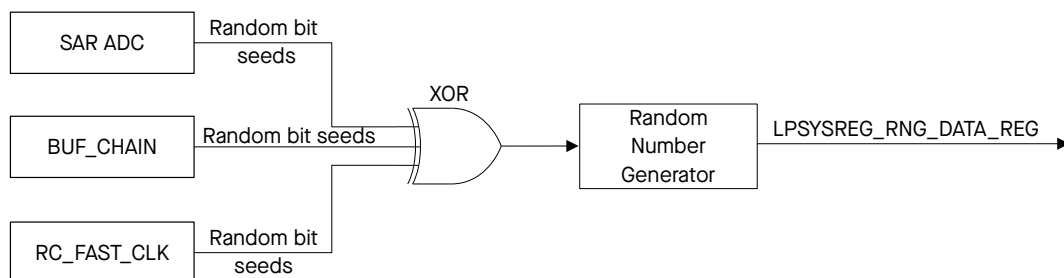


Figure 33.3-1. Noise Source

When there is noise coming from the SAR ADC, the random number generator is fed with a 1-bit entropy in one clock cycle of RC_FAST_CLK, which is generated from an internal RC oscillator (see Chapter 10 [Reset and Clock](#) for details). Thus, it is advisable to read the [LPSYSREG_RNG_DATA_REG](#) register at a maximum rate of 1 MHz to obtain the maximum entropy.

33.4 Programming Procedure

When using the random number generator, make sure at least either the SAR ADC or RC_FAST_CLK is enabled. Otherwise, pseudo-random numbers will be returned.

- For SAR ADC, please refer to Chapter 62 [ADC Controller \(ADC\)](#).
- RC_FAST_CLK is enabled by setting the PMU_HP_SLEEP_XPD_FOSC_CLK bit in the [PMU_HP_SLEEP_LP_CLK_POWER_REG](#) register. Please refer to the [Note](#) about the RC_FAST_CLK.
- RC_FAST_CLK is gated by setting the [LPPERI_CLK_EN_RNG](#) bit in the [LPPERI_CLK_EN_REG](#) register.
- The BUF_CHAIN can be enabled by setting the [RNG_CFG_REG](#) register's [RNG_SAMPLE_ENABLE](#) bit.

Note:

Enabling RC_FAST_CLK increases the RNG entropy. However, to ensure maximum entropy, it's recommended to always enable an ADC source as well.

When using the random number generator, read the [LPSYSREG_RNG_DATA_REG](#) register multiple times until sufficient random numbers have been generated. Ensure the rate at which the register is read does not exceed the frequencies described in section 33.3 above.

33.5 Register Summary

The abbreviations given in Column **Access** are explained in Section [Access Types for Registers](#).

Name	Description	Address	Access
LPSYSREG_RNG_DATA_REG	Random source data	0x5011_01A4	RO
LPPERI_CLK_EN_REG	Integrated clock gating enable signal of RC_FAST_CLK	0x5012_0000	R/W
RNG_CFG_REG	Configuration register of RNG	0x5012_6000	R/W

33.6 Registers

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 33.1. LPSYSREG_RNG_DATA_REG (0x5011_01A4)

LPSYSREG_RNG_DATA																															
31																															0
0x00000000																															
Reset																															

LPSYSREG_RNG_DATA Random number source. (RO)

Register 33.2. RNG_CFG_REG (0x5012_6000)

reserved										RNG_TIMER_EN			RNG_TIMER_PSCALE				RNG_SAMPLE			
31									10	9	8					1	0			
0x000										0			0x00				0		Reset	

RNG_SAMPLE_ENABLE Configures whether to enable the RNG asynchronous clock noise source.
1: Enable
0: Disable
(R/W)

RNG_TIMER_PSCALE Represents the prescale factor of the RNG internal timer.(R/W)

RNG_RTC_TIMER_EN Configures whether to enable the RNG internal timer.
1: Enable and use the counter value of this timer when generating random numbers.
0: Disable the timer.
(R/W)